Radioastronomy

Fortgeschrittenenpraktikum
HS1 Radioastronomy

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October 11, 2026

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1. Introduction

The experiment HS1-Radioastronomy is there to give an overview for operating a radio telescope and to introduce the prerequisites such as characterising and calibrating the telescope before carrying out the experiments. The main experiment to be conducted is the observation of neutral hydrogen in the Milky Way to determine the rotational velocity in order to derive the mass distribution.

1.1. Radioastronomy

Radio astronomy is the study of electromagnetic radiation in the radio wavelength range, typically from a few MHz to several hundred GHz. Compared to optical astronomy, radio waves are much less affected by interstellar dust and gas, which allows astronomers to observe cold and dense regions of space that are often hidden in the optical regime. Radio telescopes usually consist of large parabolic antennas that collect incoming radiation and direct it to a receiver. The receiver amplifies the weak signal and converts it into an electrical voltage, from which the intensity, spectral distribution, and temperature of the observed source can be determined.

A central quantity in radio astronomy is the brightness temperature. In the Rayleigh-Jeans regime, which is valid for most radio astronomical observations, the spectral intensity is approximately

$$I_\nu \approx \frac{2k_\mathrm{B}T_\mathrm{B}\nu^2}{c^2} = \frac{2k_\mathrm{B}T_\mathrm{B}}{\lambda^2},$$

where I_ν is the spectral intensity, k_B is the Boltzmann constant, T_B is the brightness temperature, ν is the frequency, and c is the speed of light. This relation shows that the observed radio signal is directly linked to the temperature of the emitting medium. This makes radio astronomy a powerful method for studying cold molecular clouds, ionized gas regions, and neutral hydrogen structures in the Milky Way and beyond [1].

One of the most important spectral lines in radio astronomy is the 21 cm line emitted by neutral hydrogen.

1.2. The 21 cm Hydrogen Line

The hydrogen atom consists of a proton and an electron. In the ground state, the proton and electron spins can be oriented either parallel or antiparallel to each other. This gives rise to a hyperfine splitting of the ground state: the antiparallel configuration has a slightly lower energy than the parallel configuration. A transition from the higher-energy state to the lower-energy state leads to the emission of a photon with a frequency of approximately

$$\nu \approx 1420.4 \text{ MHz},$$

which corresponds to a wavelength of

$$\lambda \approx 21.1 \text{ cm}.$$

This transition is therefore called the 21 cm line.

This line is of great importance in astronomy because hydrogen is the most abundant element in the interstellar medium. Neutral hydrogen emits strongly at 21 cm, allowing astronomers to map the distribution of gas in galaxies. The 21 cm line is especially useful because it is only weakly absorbed by interstellar dust, so it can reveal regions that are difficult or impossible to observe in the optical range [1].

The 21 cm line also enables the measurement of Doppler shifts. If the hydrogen gas is moving relative to the observer, the frequency of the emission is shifted. From this shift, the radial velocity of the gas can be determined. This makes it possible to study galactic rotation, gas dynamics, and the mass distribution of galaxies. For this reason, the 21 cm line is one of the most important tools in modern radio astronomy and is fundamental for studying neutral hydrogen in the universe.

2. Observation and Evaluation

2.1. The KRT3 Telescope

The KRT3 is a SPIDER 300A radio telescope manufactured by PrimaLuceLab SpA. It was installed in 2019 and consists of a parabolic antenna with a diameter of 3 m and a digital receiver. The antenna is mounted on an alt-azimuth mount, allowing it to move independently in elevation and azimuth and to track astronomical sources automatically.

The antenna focuses incoming radio waves onto a receiving horn located at its primary focus. A septum inside the horn separates the radiation into left-hand circular polarization (LHCP) and right-hand circular polarization (RHCP). Each polarization is amplified by a separate low-noise amplifier located directly behind the horn. This arrangement minimizes signal losses before the first amplification stage. The low-noise amplifiers provide a gain of approximately 33 dB.

The digital superheterodyne receiver covers a frequency range from 1398 MHz to 1460.5 MHz. This range includes the 21 cm hydrogen line at approximately 1420.4 MHz. The total bandwidth is divided into 1024 channels, each with a bandwidth of approximately 61 kHz. The channels are grouped into 16 Base Band Converters (BBCs): BBCs 1–8 record the left-hand circular polarization, while BBCs 9–16 record the right-hand circular polarization.

The received signals are mixed down to a lower intermediate frequency and digitized using a 14-bit analogue-to-digital converter. A digital signal processor then calculates the spectrum using a Fourier transform. The telescope and receiver are controlled with the software RadioUniversePro, which is used to point the antenna, record measurements, and display the resulting data [1].

2.2. Weather Conditions

The experiment was carried out on 07.09.2026, starting at around 12:35. The weather was stable throughout the day with partial cloud coverage and temperatures around 19°, a relative humidity of 60% and a windspeed of $1\,\mathrm{m\,s^{-1}}$.

2.3. Calibration

2.3.1. Determination of the system temperature

The receiver receives radiation not only from the source T_S but also from the environment T_h. In addition to those two external sources of radiation, the antenna also picks up on the radiation its own system emits T_{sys}. This means the measured power of the receiver, represented by the antenna temperature, can be broken down the following way:

$$T_A = T_S + T_h + T_{sys} \tag{1}$$

It is difficult to separate the pure system temperature and the background radiation, so these temperatures are usually combined into an effective system temperature:

$$T'_{sys} = T_{sys} + T_h \tag{2}$$

The effective system temperature now includes all unwanted contributions. This means that to know the contribution of a source, one must know the effective system temperature. Determining the effective system temperature is calibrating the telescope by doing a 'On-Off' measurement through measuring a source with a known flux density and then measuring a piece of empty sky. The KRT3 however, can only measure the sun as a source with high enough flux density but the sun is not a constant source which is why a noise generator needs to be used.
The noise generator injects a signal directly into the antenna horn. The calibration temperature $T_{cal} = 170\,\mathrm{K}$ is the expected antenna temperature that is observed additionally when the noise generator is switched on.
When assuming the sun to be a point-like source, the following approximation can be used:

$$T_b \Omega_s = T_A \Omega_A \tag{3}$$

with T_b being the brightness temperature, Ω_s being the solid angle of the sun, T_A being the antenna temperature and Ω_A being the solid angle of the antenna main lobe.
The goal is measuring the antenna temperature of the sun and converting it into a flux density. For this we need the following identities: the antenna theorem $A_e\Omega_A = \lambda^2$, the Rayleigh-Jeans law $B_\nu = 2kT_b/\lambda^2$, the flux density $S_\nu = B_\nu\Omega_s$ and the aperture efficiency $\eta = A_e/A_g = 0.5$ with A_g being the geometric antenna surface assumed to be a circle.

$$T_b\Omega_s = T_A\Omega_A \tag{4}$$

$$\Omega_A = \frac{\lambda^2}{A_e}, \Omega_S = \frac{S_\nu}{B_\nu}, B_\nu = \frac{2k_BT_b}{\lambda^2} \tag{5}$$

$$S_\nu = \frac{T_A \cdot 2k_B}{A_e} = f(T_A, A_e) \tag{6}$$

$$A_e = \pi r^2 \cdot \eta = 1,5^2\pi \cdot 0,5 = \pi \cdot 1.125\,\mathrm{m}^2 \tag{7}$$

$$\Rightarrow T_A = \frac{S_\nu A_e}{2k_B} = \frac{S_\nu \cdot \pi \cdot 1.125\,\mathrm{m}^2}{2k_B} \tag{8}$$

2.3.2. Background Radiation

The determination of the background radiation is done at the calibration position where $T_A \approx T'_{sys}$. With the control software of the KRT3 RadioUniversePro the easiest way to measure the background radiation is to do a cross scan over a weak source that barely stands out from the background.
For our purposes we chose the source Virgo A as it was at an elevation of 47° 56 min.

2.4. Determination of the angular resolution of the KRT3

The expected half-width is $q = 1.02\lambda/D$ with wavelength λ and antenna diameter $D = 3\,\mathrm{m}$. To determine the angular resolution the radio telescope is moved step wise over the sun and the antenna temperatures are measured.
The concrete settings in RadioUniversePro are a cross-scan in the Total Power Plots-tab with:

- Reference frame: Horizon
- Span [deg]: 20
- Step [deg]: 1
- Average [s]: 3
- Devices: BBCs
- Save as FITS

2.5. Determination of the sensitivity of the instrument

2.5.1. Theoretical sensitivity

To determine the theoretical sensitivity of the instrument, the weakest detectable flux density ΔS_ν for a given system temperature T'_{sys} can be given by the radiometer equation:

$$\Delta S_\nu = \frac{2k}{A_e} \cdot \frac{C_s T'_{sys}}{\sqrt{\Delta\nu\tau}} \tag{9}$$

where $C_s \approx 2$ is a dimensionless instrumental constant and τ the integration time.

The measurement is carried out with the radio source Vir A.
The necessary integration time for detecting Vir A with a signal-to-noise ratio of $S/N = 10$ in each channel is calcuated, taking the flux density at $\nu = 1420.4\,\mathrm{MHz}$ to be $S_\nu = 211\,\mathrm{Jy} \pm 11\,\mathrm{Jy}$ after Baars et al. (1977):

- Virgo A: Flux density = 211 ± 11 Jy, Radiogalaxy

- For 5 BBCs on each side ($\Delta\nu = 5 \cdot 61\text{kHz} = 305\text{kHz}$)
- $S/N = 10$
- $\tau = \left(\frac{2k \cdot C_S \cdot T'_{\text{sys}}}{\Delta S_\nu \cdot A_E}\right)^2 \cdot \frac{\Delta\nu}{S/N}$

2.5.2. Determination of the uncertainty of radiant flux measurements

The radio signals from space received on earth are weak. This can be illustrated by the thought experiment of receiving the signal of a mobile radio on the moon transmitting with 2 W, 900 MHz and a FWHM of 80 kHz, $\sigma = 34$ kHz being received on earth. In this scenario the flux density is:

$$S_\nu = \frac{P}{4\pi d^2 \sqrt{2\pi}\sigma} = 1263.16\,\text{Jy} \tag{10}$$

When compared with the flux density of Cas A at 900 MHz (1420 Jy ± 70 Jy), it's surprisingly high.
To calculate the flux density of Cas A, we assume the spectral response is a simple power law of $S_\nu \propto \nu^\alpha, \quad \alpha = -0.7$: $\nu = 1420.4$ MHz: $S_{\nu,\text{CasA}} = 1420 \pm 70$ Jy.
Using the previous equation $S_\nu = f(T_A, A_e)$, the antenna temperature T_A for Cas A can be calculated:

$$k = \frac{1420,4^{-0.7}\ \text{MHz}}{1420\ \text{Jy}} \tag{11}$$

$$S_{\nu,\text{Cas A}} = \frac{900^{-0.7}}{k}\text{Jy} = 1954.37\,\text{Jy} \tag{12}$$

$$T_A = \frac{S_\nu A_e}{2k_B} = 1.82\,\text{K} \quad (\text{für } \nu = 1420.4\ \text{MHz}) \tag{13}$$

To measure the flux density means to measure the source (On) and then to do a measurements without the source (Off). As we need to separate the background from the antenna temperature, a so-called Beamswitch measurement needs to carried out in which the position of the source and a reference position next to the source are measured alternately.
From this On-Off measurement the temperature of the source T_s can be determined the following way:

$$T_A(ON) - T_A(OFF) = (T_s + T'_{sys}) - T'_{sys} \tag{14}$$

The source Vir A was chosen to be observed in a grid pattern on the sky where the KRT3 takes measurements line by line at different positions. This means different columns are scanned to assemble the grid pattern into a map in the shape of pixel = rows × columns. The reason Vir A was chosen and not one of the brighter sources, was that Vir A was the only source that fit the parameters of an elevation between 30° and 60°.
The BBCs 3 and 11 were chosen as they showed the smallest RFI influence even though they show the H1 line.
The measurements were carried out in the tab Mapping with the following settings:

- Width: 10 deg; Height: 10 deg
- Delta: 0.5 HPBW
- Average: 10 s

The mapping process took 28:35.4 s according to the software and 29:27.8 s according to our own timer.

2.6. Observation of neutral hydrogen in the Milky Way

2.6.1. Determine the frequency settings of the receiver

The KRT3 has a frequency range of 1398.00 MHz to 1460.56 MHz divided up into 1024 channels with a preset bandwidth of 8 MHz.

With the bandwidth of a single channel $\delta\nu = \frac{\Delta\nu}{1024} = \frac{1460.56\,\text{MHz} - 1398.00\,\text{MHz}}{1024} = 62.56\,\text{MHz}$ and the lab frequency of the 21 cm line being $\nu_0 = 1420.4\,\text{MHz}$, the velocity resolution can be calculated:

$$\frac{\delta v}{c} = \frac{\delta\nu}{\nu_0} \tag{15}$$

$$\Leftrightarrow \delta v = \frac{c}{\nu_0} \cdot \delta\nu = 12.89\,\text{km/s} \tag{16}$$

To measure the H1 line at the lab frequency and a radial velocity of 0 km/s the channels BBC 3 and 11 can be used. Even with an assumed maximal radial velocity of ±200 km/s those channels are sufficient as the frequency shift due to the Doppler effect is only 0.94 MHz and the BBCs have a frequency range of 8 MHz.

2.6.2. Measurement of the 21cm line along the Milky Way plane

One of the approximations determining the radial velocity is the assumption that the hydrogen is distributed in the plane on circular orbits around the galactic centre. This approximation leads to the following orbital velocity:

$$v(R)^2 = \frac{GM_R}{R} \tag{17}$$

With M_R being the mass of the Milky Way within the radius R and the gravitational constant $G = 6.67 \times 10^{-11}\,\text{m}^3/\text{kg}/\text{s}^2$.

As we observe from a point in the same system we want to measure, we need to think about how to actually measure the variable R. We observe along a line of sight defined through the relative angle l with respect to the line connecting the sun and the galactic centre (see Figure 1). This means our calculation for R, the distance to the point with the highest radial velocity, is $R = R_0 \sin(l)$ with $R_0 = 8.5\,\text{kpc}$ being the distance of the sun from the galactic centre.

The orbital velocity at the tangential point is $v = \omega R$ with ω being the angular velocity at distance R.

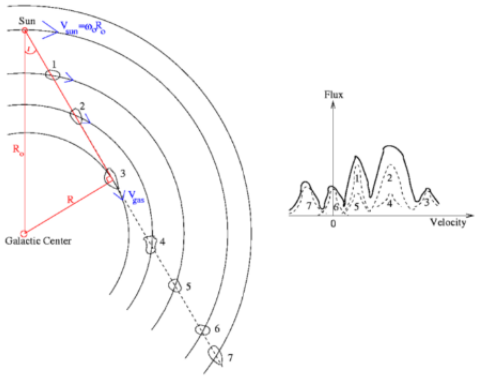


Figure 1: A schematic representation of the rotation of H1 clouds around the galactic centre together with the resulting line profile to the right [1].

This leads to the radial velocity between the tangent point and the sun:

$$v_{\text{rad}} = \omega(R)R - \omega_0 R_0 \sin(l) \tag{18}$$

$$= v(R) - \omega_0 R \tag{19}$$

The measurement carried out is an On-Off measurement at a point on the galactic plane. The location of the galactic centre is at $l = 0°$, $b = 0°$ with l being the galactic longitude and b being the galactic latitude. The spectra were taken with constant b and varied l in the range of 10° to 90°.

For that, the telescope is aligned to the galactic plane using the command line in RadioUniversePro. The USBs 3 and 11 in the BBC Tools tab were selected and the following settings were used:

- Repetitions: 10
- Average [s]: 10
- Elev cutoff [deg]: 60
- Distance [beam]: 4
- galoff
- Aquire OnOff Spectra
- Save as FITS

3. Results

3.1. Determination of the system temperature

The system temperature was determined from the calibration measurements using the relation

$$T'_{\mathrm{sys}} = T_{\mathrm{cal}} \frac{P_{\mathrm{off}}}{P_{\mathrm{cal}} - P_{\mathrm{off}}},$$

where T_{cal} is the calibration temperature and P_{off} and P_{cal} are the received powers with the noise source switched off and on, respectively. The resulting values were averaged separately for the left circular polarization (BBC 1–8) and the right circular polarization (BBC 9–16), yielding

$$T'_{\mathrm{sys,left}} = 98.76 \pm 22.54\,\mathrm{K}, \qquad T'_{\mathrm{sys,right}} = 107.80 \pm 31.65\,\mathrm{K}.$$

Using these system temperatures, the antenna temperature of the Sun was then calculated from the corresponding calibration data, giving

$$T_{\mathrm{sun,left}} = 1237.79 \pm 89.62\,\mathrm{K}, \qquad T_{\mathrm{sun,right}} = 1302.59 \pm 119.16\,\mathrm{K}.$$

The larger uncertainty in the right polarization reflects the larger scatter of the corresponding measurements. With Equation 13 and the current measured flux densities provided by the SWC of the Australian Government ($S_\nu \approxeq 76\,\mathrm{sfu}$) the expected antenna temperature was estimated to be

$$T_A = 972.75\,\mathrm{K}.$$

This means $T_{\mathrm{sun,left}}$ and $T_{\mathrm{sun,right}}$ both diverge from T_A by almost 3σ with the divergence for $T_{\mathrm{sun,left}}$ being marginally higher. This is a serious difference in determined and expected temperature and can most likely be traced back to RFI in the measurements.

3.2. Background Radiation

In order to convert the raw TPI signal into the actual background temperature, the measured data was normalized using the mean off-source power and the effective system temperature according to

$$T'_{\mathrm{h,n}} = \frac{\mathrm{TPI} \cdot T'_{\mathrm{sys}}}{\bar{P}_{\mathrm{off}}}. \tag{20}$$

Here, T'_{sys} is the effective system temperature and $\bar{P}_{\mathrm{off}}$ is the mean reference power measured during the off-source observation. This normalization removes the dependence on the receiver gain and instantaneous power level and rescales the measured TPI signal into an effective temperature. The resulting background curve (see Figure 2) therefore represents the true elevation-dependent contribution of the environment to the antenna temperature, which must be corrected for in order to recover the source signal reliably.

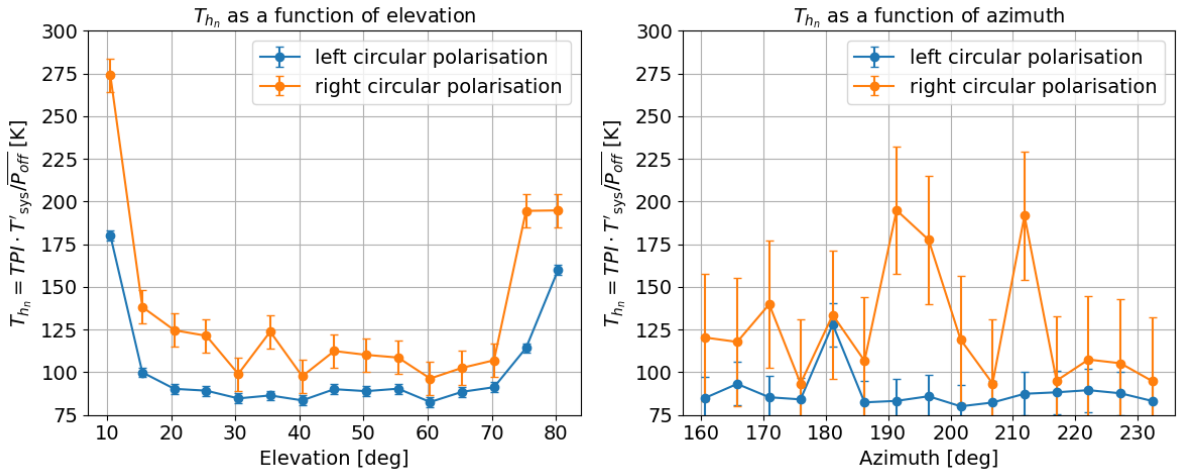


Figure 2: Effective background temperature as a function elevation (left figure) or azimuth (right figure).

The background temperature T_h is not constant, but varies significantly with elevation. The background plot shows a clear increase toward lower elevations, indicating that the antenna receives additional radiation from the atmosphere and surrounding environment when observing near the horizon. At higher elevations, the environmental contribution decreases and T_h approaches a lower value set mainly by sky brightness and instrumental noise.

The background temperature also varies with azimuth, showing that the local environment contributes differently depending on the telescope pointing direction. This is consistent with differing amounts of ground spillover and thermal emission from nearby structures in different azimuthal directions.

The uncertainty is significantly larger for the right-hand polarization than for the left-hand polarization, especially as a function of azimuth. This suggests that the right channel is more strongly affected by instrumental systematics and environmental variations. The smaller uncertainties in the left polarization indicate a more stable response and a more reliable background determination.

Using the extrapolated course of T'_h (see Figure 3) up to an elevation of 0°, it becomes clear that the background temperature rises strongly at low elevations. A comparison with the Earth's effective radiation temperature, $T_{\text{Earth}} \approx 252\,\text{K}$, shows that this increase is of a comparable magnitude and therefore reflects the dominance of atmospheric and ground emission at low elevation.

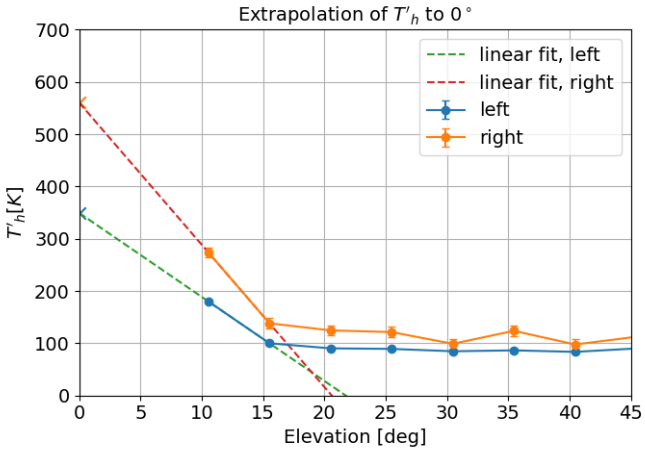


Figure 3: The extrapolated course of T'_h up to an elevation of 0° (for left and right hand polarisation)

3.3. Determination of the Angular Resolution of the KRT3

The angular resolution of the KRT3 was determined by scanning the Sun in longitude and latitude. The measured signal was normalized to the expected solar antenna temperature $T_{A,\mathrm{Sun}}$. The resulting beam profiles were fitted with Gaussian functions, as shown in Figures 4 and 5. The angular resolution was obtained from the full width at half maximum (FWHM) of the fitted Gaussian profiles.

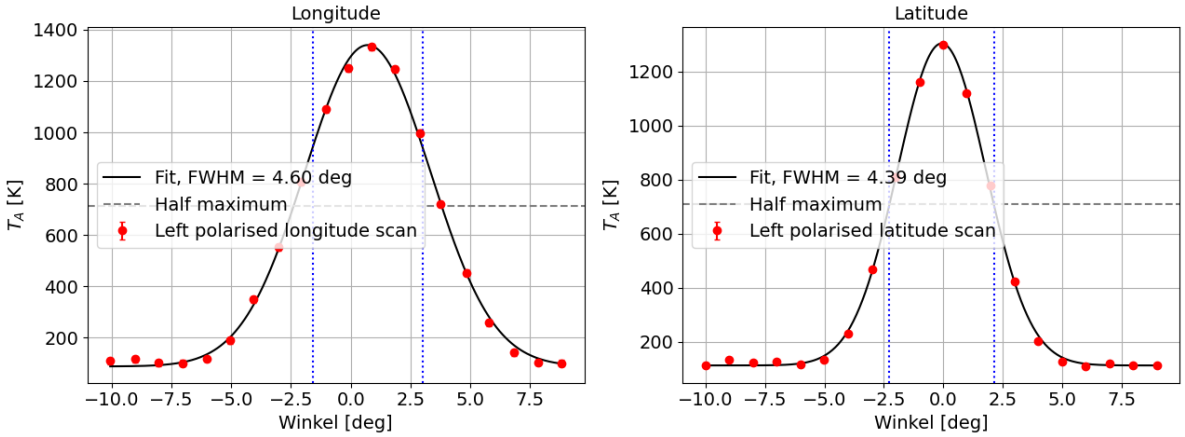


Figure 4: Gaussian fit of the left polarised scan of the longitude (left figure) and latitude (right figure). The FWHM of the fit is marked at 4.6° for the longitude scan and at 4.39° for the latitude scan. The errorbars in both figures are to small to be seen at the scale of the antenna temperature.

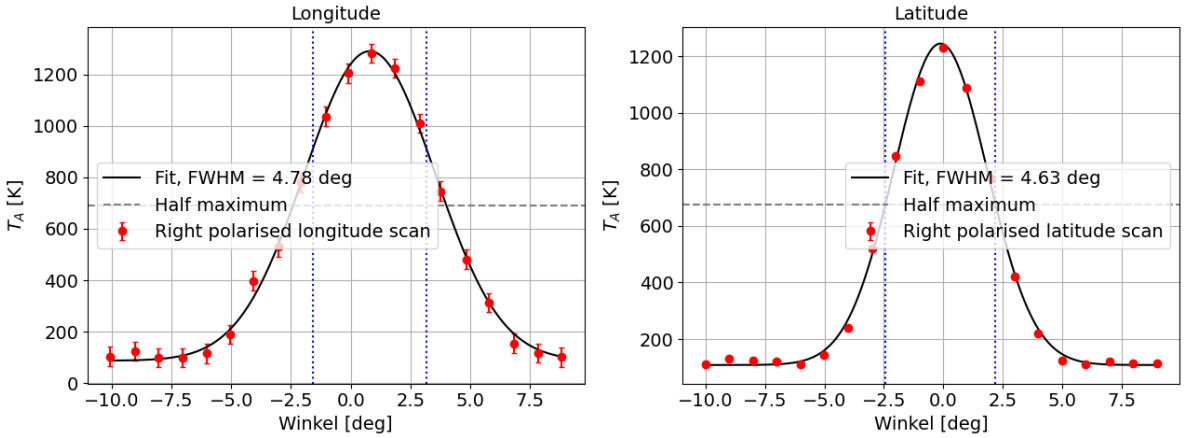


Figure 5: Gaussian fit of the right polarised scan of the longitude (left figure) and latitude (right figure). The FWHM of the fit is marked at 4.78° for the longitude scan and at 4.63° for the latitude scan. The errorbars in the right figure are remarkably high when contrasted with both the left figure as well as Figure 4.

The resolution was determined separately for the longitude and latitude scans and for both circular polarization channels. In the longitude scans, the measured angular offset does not correspond directly to a great-circle distance on the sky. Therefore, the angular width was corrected using the factor cos(ELV), where ELV is the elevation of the Sun (ELV = 42°):

$$\theta_{\mathrm{lon}} = \mathrm{FWHM}_{\mathrm{lon}} \cos(\mathrm{ELV}).$$

For the latitude scans, no such correction is required because the scan follows a great circle approximately directly in elevation.

The theoretical angular resolution of the KRT3 is estimated using

$$\theta_{\text{theo}} \approx 1.02 \frac{\lambda}{D},$$

where $D = 3\,\text{m}$ is the antenna diameter and $\lambda = 21.1\,\text{cm} = 0.211\,\text{m}$ is the observing wavelength. This gives

$$\theta_{\text{theo}} \approx 1.02 \frac{0.211}{3} \approx 0.0717\,\text{rad} \approx 4.11^\circ.$$

The measured resolutions are slightly larger than this value for all scans. For the longitude scan, the measured values are

$$\theta_{\text{lon,left}} = 4.60^\circ \pm 0.06^\circ, \qquad \theta_{\text{lon,right}} = 4.78^\circ \pm 0.08^\circ,$$

which are larger than the theoretical value by approximately 11.9% and 16.3%, respectively. For the latitude scan, the measured resolutions are

$$\theta_{\text{lat,left}} = 4.39^\circ \pm 0.04^\circ, \qquad \theta_{\text{lat,right}} = 4.63^\circ \pm 0.07^\circ,$$

corresponding to deviations of approximately 6.8% and 12.7%.

The measured values are therefore in reasonable agreement with the theoretical resolution. The small systematic increase can be explained by pointing uncertainties, limited sampling, atmospheric fluctuations, imperfections in the Gaussian fit, and the finite angular diameter of the Sun. The comparatively large uncertainties result from the scatter of the measurement points and the limited number of data points used for the Gaussian fits.

3.4. Determination of the Sensitivity of the instrument

3.4.1. Theoretical sensitivity

With the system temperatures $T'_{\text{sys, left}} = 98.76\,\text{K} \pm 22.54\,\text{K}$ and $T'_{\text{sys, right}} = 107.80\,\text{K} \pm 31.65\,\text{K}$ determined in subsection 3.1 together with Equation 9, the limiting sensitivity in a single channel can be calculated to:

$$\Delta S_{\nu,\text{left}} = 6,248 \cdot 10^{-24} \frac{\text{W}}{\text{m}^2\text{Hz}} = 624.92\,\text{Jy} \tag{21}$$

$$\Delta S_{\nu,\text{right}} = 6,82 \cdot 10^{-24} \frac{\text{W}}{\text{m}^2\text{Hz}} = 682.02\,\text{Jy} \tag{22}$$

With $A_e = 1.125\pi$, $\tau = 1\,\text{s}$ and $\Delta\nu = 61\,\text{kHz}$.
The determined system temperatures can now also be used to calculate the necessary integration time for detecting Vir A using Equation 2.5.1: $\tau_l = 4,12\text{s}$, $\tau_r = 5,22\text{s}$.

3.4.2. Determination of the uncertainty of radiant flux measurements

The map was evaluated using SAOImageDS9. A region containing 35 pixels was selected away from the source and from the visible disturbance (see Figure 7). In the uncalibrated map units, the mean and standard deviation were

$$\overline{P}_{\text{off}} = 1455.83, \tag{23}$$

$$\sigma_P = 212.206. \tag{24}$$

The map uses BBC 11, i.e. the right-hand polarization, for which $T'_{\text{sys}} = 107.80\,\text{K}$. Each map value is converted using the same scale factor, $T'_{\text{sys}}/\overline{P}_{\text{off}}$. Thus,

$$T_A(\text{OFF}) = \frac{\overline{P}_{\text{off}}}{\overline{P}_{\text{off}}} T'_{\text{sys}} = 107.80\,\text{K}, \tag{25}$$

$$\sigma_T = \frac{\sigma_P}{\overline{P}_{\text{off}}} T'_{\text{sys}} = 15.71\,\text{K}. \tag{26}$$

Applying this scale to the measured map value at the position of Virgo A gives

$$T_A(\mathrm{ON}) = 227.10\,\mathrm{K}. \tag{27}$$

As a result, using the on-off relation from Equation 14, the antenna temperature of the source is

$$T_s = T_A(\mathrm{ON}) - T_A(\mathrm{OFF}) = 119.30\,\mathrm{K}. \tag{28}$$

This on-off difference is not consistent with Virgo A. For the flux density $S_\nu = 211\,\mathrm{Jy} \pm 11\,\mathrm{Jy}$ at 1420.4 MHz given above and $A_e = 1.125\pi\,\mathrm{m}^2$, the theoretical antenna temperature is

$$T_{A,\mathrm{Vir\,A}} = \frac{S_\nu A_e}{2k_B} = 0.270\,\mathrm{K} \pm 0.014\,\mathrm{K}. \tag{29}$$

The measured map scatter of $\sigma_T = 15.71\,\mathrm{K}$ is about 58 times larger than this expected signal; the theoretical signal-to-noise ratio is only about 0.017. The apparent measured difference of 119.30 K is roughly 440 times the theoretical antenna temperature and therefore cannot be attributed to Virgo A. The map does not provide a credible detection; the large excess is likely caused by interference or other measurement systematics. The quoted map scatter describes spatial variations and does not include the uncertainty of the system-temperature calibration.

The complete mapping of the 7×7 map required about 28 min 35.4 s according to the software, or 29 min 27.8 s according to the independent timer. With an integration time of 10 s per measurement point, Virgo A fell into one pixel and was therefore observed for only 10 s. This corresponds to $\approx 0.58\%$ of the total mapping time.

Assuming $S/N \propto \sqrt{\tau}$, the number of equivalent maps needed for $S/N = 10$ is $N_{10} = (10/(S/N))^2$. The theoretical Virgo A signal gives $S/N_{\mathrm{theo}} \approx 0.017$ and $N_{10,\mathrm{theo}} \approx 3.4 \times 10^5$. Using the apparent map excess instead gives

$$(S/N)_{\mathrm{meas}} = \frac{119.30\,\mathrm{K}}{15.71\,\mathrm{K}} \approx 7.6, \qquad N_{10,\mathrm{meas}} = \left(\frac{10}{(S/N)_{\mathrm{meas}}}\right)^2 \approx 1.7. \tag{30}$$

The measured value is only an apparent map S/N, not a Virgo A detection: the excess is inconsistent with the theoretical signal, and σ_T measures spatial variation rather than repeated-measurement noise.

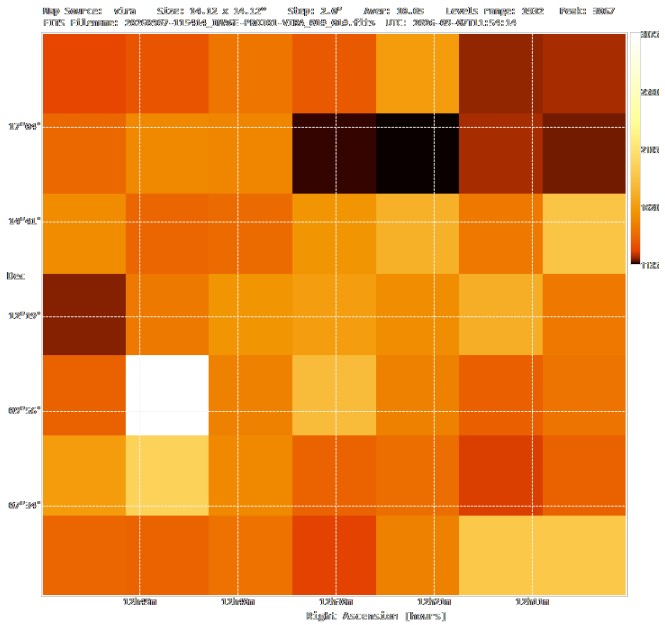


Figure 6: Map of the galaxy Virgo A

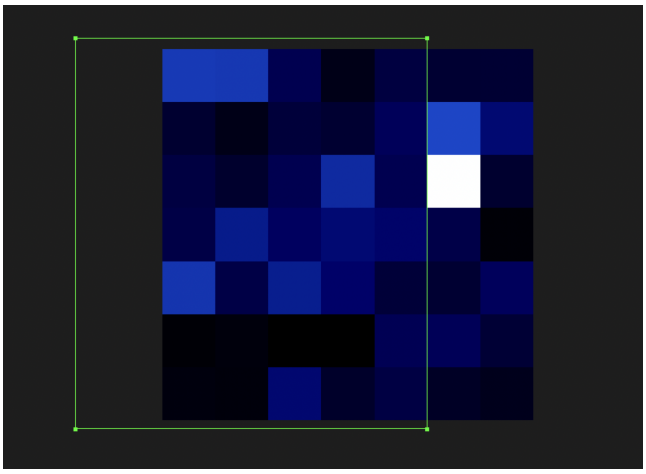


Figure 7: The selected region for the mean of the map, avoiding the probable disturbance.

3.5. Observation of neutral hydrogen in the Milky Way

The measured spectra for the varying galactic longitudes were averaged using the provided python script SPECTRUM.PY.
When comparing the spectrum for $l = 80°$ with the H I-line profile of the Leiden-Argentine-Bonn (LAB) survey, as done in Figure 8, we can see that our measured data matches the three-fingered shape of the LAB spectrum roughly with a few differences. Those differences being a slight red-shift, the middle peak being slightly higher and the right peak being about 20 K higher.

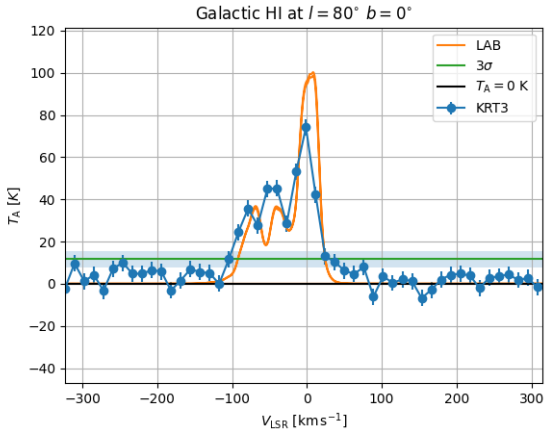


Figure 8: The measured 21 cm-line spectrum for $l = 80°$ overlaid with the LAB data of the spectrum for the same position.

For each of the averaged spectra the channel with the maximum radial velocity of the H I-profile was determined. This was done graphically, by taking the intercept of the spectrum with the 3σ-line.

l ±2 [deg]	R [kpc]	v_{max} [km s^{-1}]	v_R [km s^{-1}]
10	1.48 ± 4.09	–	38.31 ± 4.09
20	2.91 ± 4.00	70.70 ± 14.50	146.02 ± 15.04
30	4.25 ± 3.84	89.23 ± 16.67	199.23 ± 17.11
40	5.46 ± 3.61	71.20 ± 5.38	212.52 ± 6.48
50	6.51 ± 3.31	69.26 ± 4.07	237.75 ± 5.25
60	7.36 ± 2.92	30.67 ± 3.87	221.16 ± 4.85
70	7.99 ± 2.41	19.89 ± 1.01	226.69 ± 2.61
80	8.37 ± 1.72	30.95 ± 11.00	247.59 ± 11.13
90	8.50 ± 0.00	15.85 ± 3.83	235.85 ± 3.83

Table 1: The values of R, v_{max} and v_R for each galactic longitude measured.

Now the orbital velocities at the tangential points are calculated using Equation 19: Table **??**.

When plotting the rotation curve against the distance from the galactic center (see Figure 9), the expected form of the curve is a constant line, however we can clearly see a rise in velocity for bigger distances.

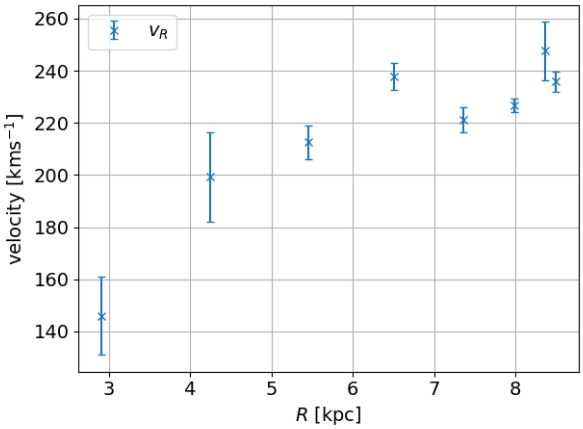


Figure 9: The radial velocity at each distance R.

Now we calculate the enclosed mass M_R of the milky way using:

$$M_R = 0.234 \times 10^{10} \left(\frac{v(R)}{100\,\mathrm{km/s}}\right)^2 \left(\frac{R}{\mathrm{kpc}}\right) M_{\mathrm{sun}} \tag{31}$$

with 1 $\mathrm{M_{sun}} = 1.98 \times 10^{30}$ kg being one solar mass. When plotting the enclosed mass (see Figure 10), a steady rise of mass can be observed. The discrepancy with the expected distribution for the visible matter (a thin disc), can be explained with the existence of dark matter.

To separate the baryonic from the dark matter in both distributions, we need to calculate the orbital velocity of the baryonic matter using the approximation of the shape as a disc:

$$v_d = 452\sqrt{\frac{\mathrm{kpc}}{R}\left[1 - \left(\frac{R}{2\mathrm{kpc}}\right)\right]}\,\mathrm{kms}^{-1} \tag{32}$$

With the distinction between baryonic and dark matter in our distributions (see Figure 11 and Figure 12) we can clearly see that while baryonic matter is distributed approximately evenly across the galaxy, dark matter bunches at the outskirts of the milky way. The mass of dark matter enclosed within the solar orbit is $M_{\mathrm{DM}}(R_0) \approx 66.41 \times 10^9 \pm 3.59 \times 10^9\, M_\odot$.

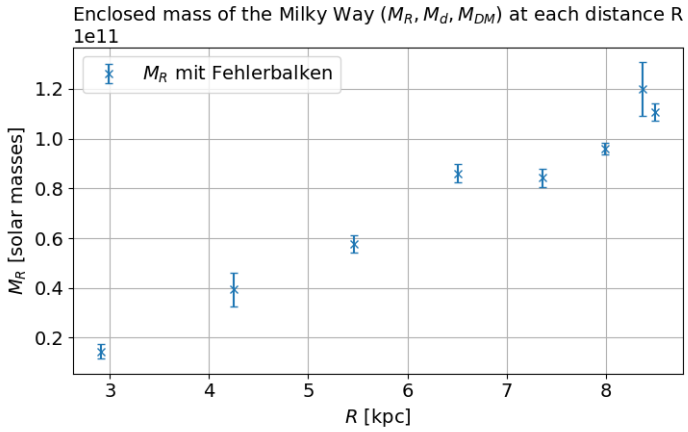


Figure 10: The enclosed mass of the milky way at each distance R.

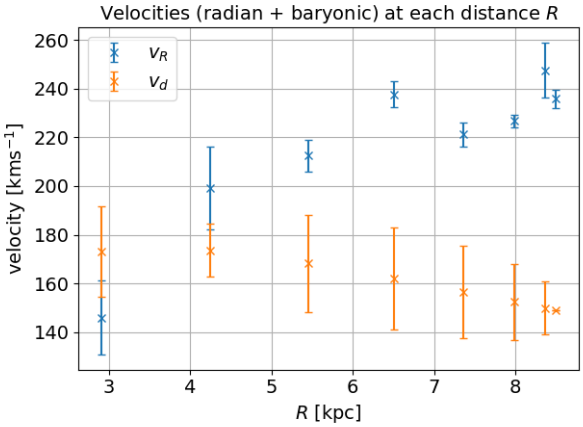


Figure 11: The radial and baryonic velocities at each distance R.

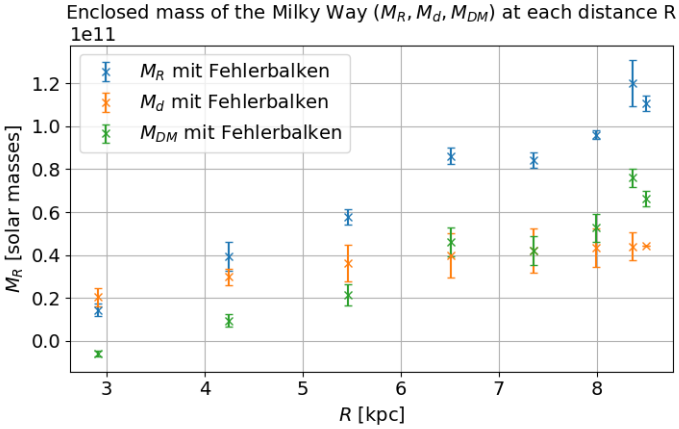


Figure 12: The enclosed mass of the milky way at each distance R broken down for the different mass distributions with M_R being the total enclosed mass, M_d being the enclosed baryonic mass and M_{DM} being the enclosed mass of the dark matter.

Assuming a spherical distribution for the dark matter $M_{DM}(R_0) = \frac{4}{3}\pi\langle\rho_{DM}\rangle R_0^3$, the mean density $\langle\rho_{DM}\rangle c^2 = 0.98\,\mathrm{GeV/cm^3} \pm 0.05\,\mathrm{GeV/cm^3}$ of the dark matter within the solar orbit can be calculated.

4. Discussion

The main objectives of the experiment were to characterise the KRT3 radio telescope, determine its angular resolution and sensitivity, and use the 21-cm line to investigate the rotation of the Milky Way. Overall, these objectives were achieved, although several measurements were affected by systematic uncertainties and external influences.

The system temperatures were determined separately for the two circular polarization channels. The left-hand polarization yielded $T'_{sys,left} = 98.76 \pm 22.54K$, while the right-hand polarization yielded $T'_{sys,right} = 107.80 \pm 31.65K$. The larger uncertainty in the right-hand channel indicates a less stable receiver response or stronger interference in this polarization. The measured solar antenna temperatures were somewhat higher than the theoretical value of approximately 972.75 K. This difference can be caused by radio-frequency interference, calibration uncertainties, and variations in the solar flux during the measurement.
The background radiation showed a clear dependence on elevation and azimuth. The background temperature increased at low elevations because the antenna received more atmospheric emission, ground radiation, and radiation from nearby structures. The azimuthal variation indicates that the local environment was not uniform around the telescope.

The measured angular resolutions were close to the theoretical value of $\leq 4.11°$. The measured values were between 4.39° and 4.78°, and were therefore slightly larger than expected. This agreement is reasonable considering the positioning accuracy of the telescope, the finite angular diameter of the Sun, atmospheric fluctuations, limited sampling, and uncertainties in the Gaussian fits. The slightly larger values obtained for the right-hand polarization are consistent with its greater measurement uncertainty.

The map does not provide a credible detection of Virgo A. Its theoretical antenna temperature, approximately 0.270 K for a flux density of 211 Jy, is far below the measured map scatter of 15.71 K. The much larger apparent on-off difference is inconsistent with the expected source signal and is most likely caused by radio-frequency interference, receiver instabilities, background variations, or other measurement systematics. Consequently, the ideal relation $S/N \propto \sqrt{\tau}$ cannot be used to claim a detection from this map.

The 21-cm measurements successfully revealed the spectral line of neutral hydrogen at different galactic longitudes. The measured spectra showed the expected changes in the radial velocity of the hydrogen gas. However, the derived rotation curve deviated from the ideal expectation of a nearly constant orbital velocity. The uncertainties in identifying the line maximum, the limited frequency resolution, baseline variations, and the relatively small number of measurements contribute to this deviation.
The determined value of the mean density of dark matter within the solar orbit was calculated to be $\langle\rho_{DM}\rangle c^2 = 0.98\,\mathrm{GeV/cm^3} \pm 0.05\,\mathrm{GeV/cm^3}$, a slightly higher value than the literature value of $\langle\rho_{DM}\rangle c^2 = 0.43\,\mathrm{GeV/cm^3} \pm 0.11\,\mathrm{GeV/cm^3}$. This difference stems from the fact, that we determined the mean value inside the path of the sun rather than the value at the radius of the sun, like the literature value is.
The choice of baseline is particularly important for the H I measurements. The baseline represents the continuum level and instrumental background, which must be subtracted before determining the line intensity and its maximum velocity. If the baseline is chosen incorrectly, the position of the line maximum and therefore the derived radial velocity can be shifted. This effect is especially relevant for weak or broad spectral features and can produce systematic errors in the rotation curve. In addition, radio-frequency interference and residual background structure may distort the baseline. Despite these limitations, the measurements provide a qualitatively consistent demonstration of the distribution and kinematics of neutral hydrogen in the Milky Way.

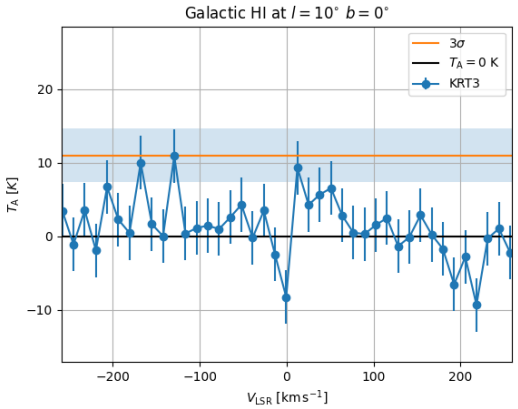
References

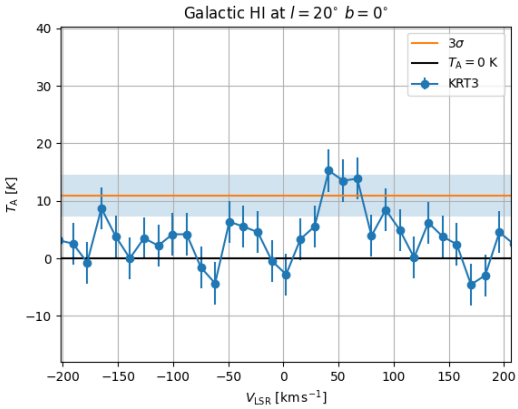
[1] D. E. V. Heesen, G. Lusetti, “Hs1 radioastronomy,” 05/05/2025.

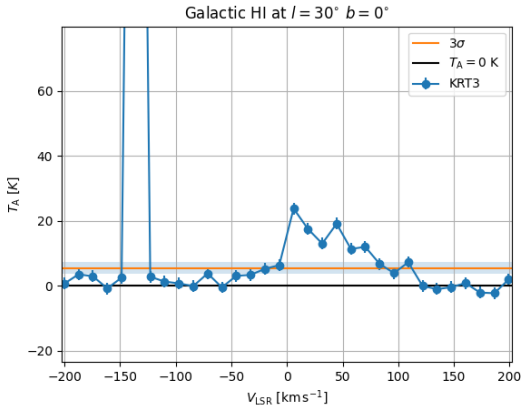
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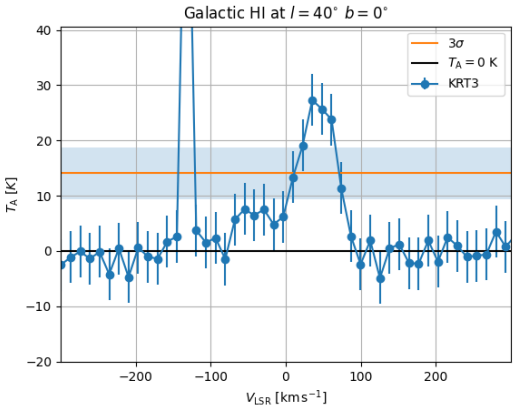
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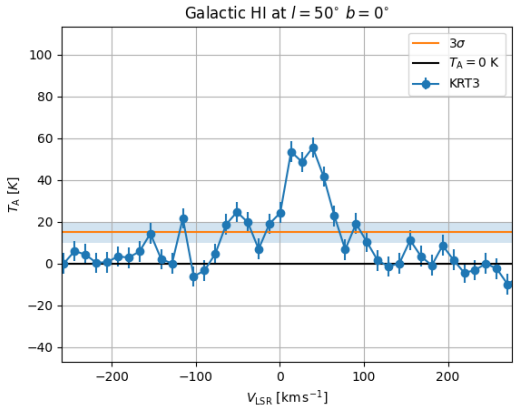
A. H_1 spectra

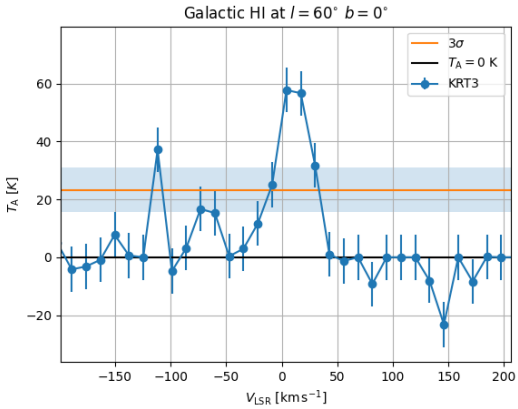

Galactic HI at $l = 10^\circ$ $b = 0^\circ$
3σ
$T_A = 0$ K
KRT3
T_A [K]
V_{LSR} [kms^{-1}]

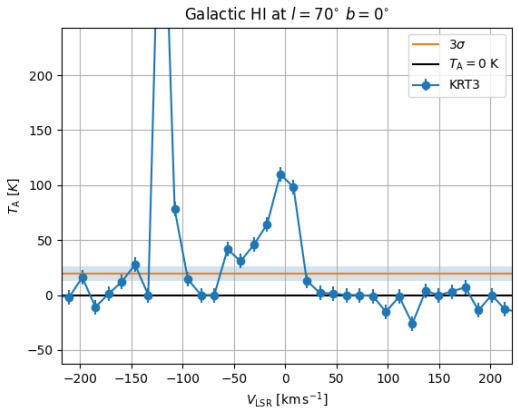

Galactic HI at $l = 20^\circ$ $b = 0^\circ$
3σ
$T_A = 0$ K
KRT3
T_A [K]
V_{LSR} [kms^{-1}]

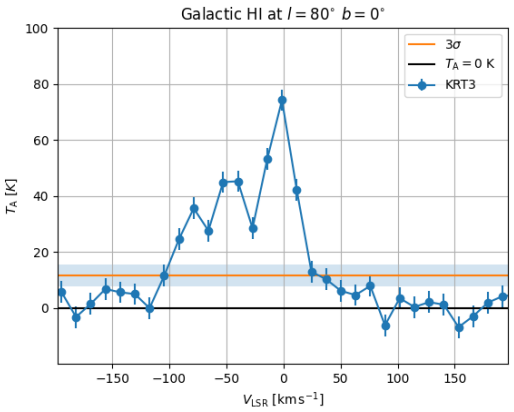

Galactic HI at $l = 30^\circ$ $b = 0^\circ$
3σ
$T_A = 0$ K
KRT3
T_A [K]
V_{LSR} [kms^{-1}]

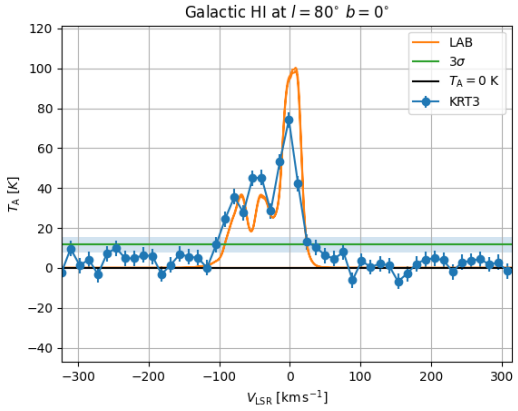

Galactic HI at $l = 40^\circ$ $b = 0^\circ$
3σ
$T_A = 0$ K
KRT3
T_A [K]
V_{LSR} [kms^{-1}]

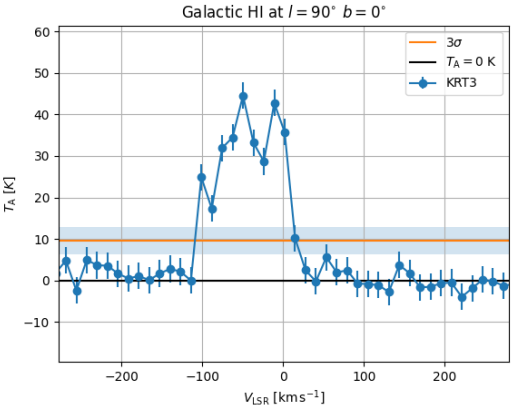

Galactic HI at $l = 50^\circ$ $b = 0^\circ$
3σ
$T_A = 0$ K
KRT3
T_A [K]
V_{LSR} [kms^{-1}]


Galactic HI at $l = 60^\circ$ $b = 0^\circ$
3σ
$T_A = 0$ K
KRT3
T_A [K]
V_{LSR} [kms^{-1}]


Galactic HI at $l = 70^\circ$ $b = 0^\circ$
3σ
$T_A = 0$ K
KRT3
T_A [K]
V_{LSR} [kms^{-1}]


Galactic HI at $l = 80^\circ$ $b = 0^\circ$
3σ
$T_A = 0$ K
KRT3
T_A [K]
V_{LSR} [kms^{-1}]


Galactic HI at $l = 80^\circ$ $b = 0^\circ$
LAB
3σ
$T_A = 0$ K
KRT3
T_A [K]
V_{LSR} [kms^{-1}]


Galactic HI at $l = 90^\circ$ $b = 0^\circ$
3σ
$T_A = 0$ K
KRT3
T_A [K]
V_{LSR} [kms^{-1}]

Labbook

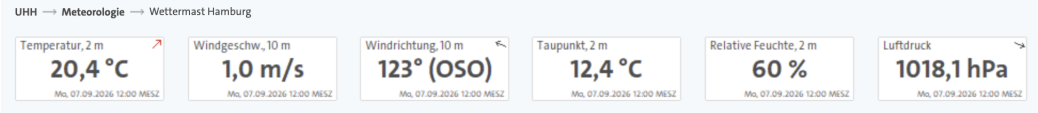
Fortgeschrittenenpraktikum
HS1 Radioastronomy

Léonie Lanz
Nico Kalogerakis

07. - 09. September 2026

Wetter

Datum: 07/09/26
Uhrzeit: 12:35 Uhr



- Teilweise bedeckter Himmel
- Kein Niederschlag

5.1.1 Calibration (Systemtemperatur)

USB 1-5, 9-13
Datendatei-Struktur im Programm (RadioUniversePro):

rep	dev	sb	on_tbi	off_tbi	cal_tbi	flux	SEFD
1	bbc01	u	12195,30	1001,40	2634,20	00,00	00,00

$$T_b \cdot \Omega_S = T_A \Omega_A \tag{33}$$

$$\Omega_A = \frac{\lambda^2}{A_e}, \quad \Omega_S = \frac{S_\nu}{B_\nu}, \quad B_\nu = \frac{2k_B T_b}{\lambda^2} \tag{34}$$

$$S_\nu = \frac{T_A \cdot 2k_B}{A_e} = f(T_A, A_e) \tag{35}$$

$$A_e = \pi r^2 \cdot \eta = 1,5^2 \pi \cdot 0,5 = 1,125\pi\,\mathrm{m}^2 \tag{36}$$

$$T_A = \frac{S_\nu A_e}{2k_B} \tag{37}$$

$T_{\mathrm{sys,left}} = 98.76 \pm 22.54$ K
$T_{\mathrm{sys,right}} = 107.80 \pm 31.65$ K
$T_{\mathrm{sun,left}} = 1237.79 \pm 89.62$ K
$T_{\mathrm{sun,right}} = 1302.59 \pm 119.16$ K

Mit S_ν der Sonne (aus SWC):

- 76 sfu (solar flux units): $76 \cdot 10^{-11}\,\mathrm{Wm^{-2}Hz^{-1}} = 76 \cdot 10000\,\mathrm{Jy} = 760000\,\mathrm{Jy}$
- $\Rightarrow T_A = 972,753$ K
- Störung wegen RFI

5.2 Angular Resolution

USB 1-5, 9-13
Position der Sonne berechnet durch die Steuerungssoftware:

Source	RA	DEC	AZ	EL	HA	Type	Note
Sun	110410.1	+055810.9	+1684733.7	+42 00 48.8	-003253.9	Star	East

- Im Ordner `cscan` befindet sich der Screenshot.
- Im Ordner `FITS` befinden sich die Daten (lon vs. lat).

5.3 Sensitivity of the Instrument

Antennentemperatur von VirA (BBC 3,11 zeigt geringsten RFI-Einfluss).
Benötigte Zeit für das Mapping:

- 28:35.4s (Software)
- 29:27.8s (Eigener Timer)

Hinweis: Screenshot auf Ubuntu und nicht Windows! Files im Ordner `maps` und in `FITS` als `imageproJ01 VirA 01`.

TODO: Theoretical Sensitivity

$$\Delta S_\nu = \frac{2k}{A_e}\frac{C_S T'_{\mathrm{sys}}}{\sqrt{\Delta\nu\tau}} \quad \text{(spektrale Flussdichte)} \tag{38}$$

Mit $C_S \approx 2$, $T'_{\mathrm{sys,left}} = 98.76 \pm 22.54$ K, $T'_{\mathrm{sys,right}} = 107.80 \pm 31.65$ K, $A_e = 1.125\pi$, $\tau = 1$s, $\Delta\nu = 61$kHz:

$$\Delta S_{\nu,\mathrm{left}} = 6,248 \cdot 10^{-24}\frac{\mathrm{W}}{\mathrm{m^2Hz}} = 624,92\ \mathrm{Jy} \tag{39}$$

$$\Delta S_{\nu,\mathrm{right}} = 6,82 \cdot 10^{-24}\frac{\mathrm{W}}{\mathrm{m^2Hz}} = 682,02\ \mathrm{Jy} \tag{40}$$

Benötigte Integrationszeit τ zum Detektieren der ausgewählten Quelle mit dem KRT 3 mit einem Signal-Rausch-Verhältnis $S/N = 10$ in jedem Kanal:

- Virgo A: Flussdichte $= 211 \pm 11$ Jy, Radiogalaxie
- Für 5 Kanäle jede Seite ($\Delta\nu = 5 \cdot 61\mathrm{kHz} = 305\mathrm{kHz}$)
- $S/N = 10$
- Formel: $\tau = \left(\frac{2k \cdot C_S \cdot T'_{\mathrm{sys}}}{\Delta S_\nu \cdot A_E}\right)^2 \cdot \frac{\Delta\nu}{S/N}$
- $\tau_l = 4,12$s
- $\tau_r = 5,22$s

Determination of the Uncertainty of Radiant Flux Measurements

Preliminary Work

$$S_\nu = \frac{P}{4\pi d^2 \sqrt{2\pi}\sigma} = 1263,16 \text{ Jy} \tag{41}$$

Für Mobile Radio on the Moon (übertragen mit 2W, 900MHz, FWHM: 80kHz, $\sigma = 34$kHz, empfangen auf der Erde).

$$S_\nu \propto \nu^\alpha, \quad \alpha = -0,7 \tag{42}$$

Bei $\nu = 1420.4$ MHz: $S_{\nu,\text{CasA}} = 1420 \pm 70$ Jy.

$$c = \frac{1420,4^{-0,7} \text{ MHz}}{1420 \text{ Jy}} \tag{43}$$

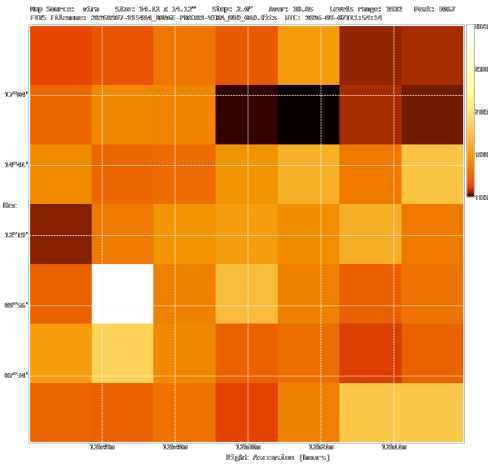
$$S_{\nu,\text{Cas A}} = \frac{900^{-0,7}}{c} \text{ Jy} = 1954,37 \text{ Jy} \tag{44}$$

$$T_A = \frac{S_\nu A_e}{2k_B} = 1,82 \text{ K} \quad (\text{für } \nu = 1420.4 \text{ MHz}) \tag{45}$$

TODO: Fehlerfortpflanzung

Evaluation

BBC 3,11



```
center=-1.8189401 0.015094938
wcs
1 pixel = 1

reg     sum             error   area            surf_bri        surf_err
                                (pix**2)        (sum/pix**2)    (sum/pix**2)
---     ---             -----   --------        ------------    ------------
1       50954           225.73  35              1455.83         6.44943

reg     sum     npix    mean    median  min     max     var     stddev  rms
---     ---     ----    ----    ------  ---     ---     ---     ------  ---
1       50954   35      1455.83 1432    1135    1896    45031.3 212.206 1471.21
```

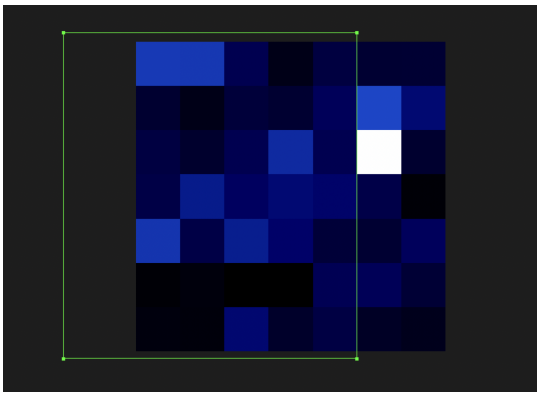
$$\text{mean} = 1455,83 \tag{46}$$

$$\text{std} = 212,206 \tag{47}$$

$$\sigma[K] = \frac{\text{std}}{\text{mean}} \cdot T'_{\text{sys}} = 15,71 \text{ K} \tag{48}$$

$$\text{mean}_{\text{scaled}}[K] = T_A(\text{OFF}) = 156938,474 \text{ K} \tag{49}$$

$$T_A(\text{ON}) = 3067 \cdot 107,8\text{K} = 330622,6 \text{ K} \tag{50}$$



Antennentemperatur der Quelle:

$$T_s = \frac{T_A(\text{ON}) - T_A(\text{OFF}) + T'_{\text{sys}}}{T'_{\text{sys}}} = 1612,17 \text{ K} \tag{51}$$

- Rauschen ist viel größer als erwartet.
- Vergleich mit theoretischer Empfindlichkeit (600 Jy, ca. 1 K).
- Bei uns jetzt 128 Kanäle, 10s Integrationszeit → eigentlich sollte es noch besser sein ($\ll 1$ K).
- Bei uns jedoch viel Einfluss von RFI →$\sim$ 15 K Std./Rauschen.
- Empfänger-/Verstärkersensibilität ist ein Problem; thermisches Rauschen begrenzt nicht, aber besser als 1 K kann es eigentlich nicht werden.

Gesucht: S/N (25 Karten für $S/N = 10$).

5.1.2 Background Radiation

Mond

Elevation: $30° 52' 32.0''$ (in Grad, Minuten, Sekunden)
Für BBC 1-5, 9-13 (wie zuvor für die Sonne, auch bei schlechter Qualität).

$$\text{Span} = 2(\text{ELV} - 10°) = 2(30° - 10°) = 40° \tag{52}$$

Da der Azimut bei $268° 03' 32.7''$ liegt, wählen wir einen etwas kleineren Span: Span $= 38°$.
Zeitpunkt: ca. 14:56 Uhr.
Daten in FITS: `LON-MOON` und `LAT-MOON` + Screenshot in `cscan`: `CSCAN-MOON`.

Virgo A

Zeitpunkt: 15:28 Uhr
Elevation: $47° 56' 18''$, Azimut: 196°, Span = 74.

Observation of Neutral Hydrogen in the Milky Way

Frequency Setting of the Receiver

Preliminary Work

- Was ist die Frequenzauflösung $\delta\nu$ (Bandbreite eines einzelnen Kanals) und wie lautet die Geschwindigkeitsauflösung δv bei der Beobachtung der H I-Linie?
- In welchem Kanal würden Sie die H I-Linie bei $\nu_0 = 1420.4$ MHz erwarten, unter Annahme einer Radialgeschwindigkeit von 0 km s^{-1}?

- Sie wollen das Radiokontinuum einer Quelle messen. Welche Frequenzen sind innerhalb der Bandbreite erlaubt, damit die Messung nicht durch die 21cm-Linie kontaminiert wird? (Maximale Radialgeschwindigkeit des H I: ± 200 km s^{-1}).
- Welche Frequenzbänder sollten in BBC Tools ausgewählt werden, wenn man einmal die H I-Linie und einmal das Kontinuum beobachten möchte?

Measurement

$$R = R_0 \cdot \sin(l), \quad R_0 = 8,5 \text{ kpc} \tag{53}$$

l [deg]	R [kpc]	$v_{\max}$ [km/s]	v_R [km/s]
10	$1,48 \pm 4,09$	–	–
20	$2,91 \pm 4,0$	$70,70 \pm 14,50$	...

- $v_{\max}$ grafisch bestimmt (Schnittwert mit der orangen σ-Linie).
- $v_R = v_{\max} + \omega_0 R_0 \cdot \sin(l)$ mit $\omega_0 R_0 = 220$ km/s.
- $v_{\max,\text{err}}$ grafisch über blauen Bereich bestimmt (Schnittpunkt min – Schnittpunkt max / 2).
- $\Delta l = \pm 2°$.
- Fehlerfortpflanzung für R:

$$\Delta R = \left|\frac{\partial R}{\partial l}\right| \cdot \Delta l = |R_0 \cdot \cos(l)| \cdot \Delta l \tag{54}$$

$$\sigma_R = \sqrt{|R_0 \cos(l)| \Delta l} \tag{55}$$

- Eingeschlossene Masse:

$$M(R) = 0,234 \cdot 10^{10} \left(\frac{v(R)}{100 \text{ km/s}}\right)^2 \left(\frac{R}{\text{kpc}}\right) M_\odot, \quad 1 M_\odot = 1,98 \cdot 10^{30} \text{ kg} \tag{56}$$

Distribution of Dark Matter

$$M_{\text{DM}}(R_0) = 66411575447.22 \pm 3593349279.00 \, M_\odot \tag{57}$$